

ENGR098 LAB 12

December 3, 2025

Objective

The purpose of this lab is to build and analyze two versions of a Wien-bridge oscillator:

- Part 1: Wien-bridge oscillator without diodes (no amplitude stabilization)
- Part 2: Wien-bridge oscillator with diodes (automatic amplitude control)

Waveforms, measured frequencies, and calculations are documented below.

Theory

A Wien-bridge oscillator uses a lead-lag RC network as its frequency-selective feedback network. The oscillation frequency is:

$$f = \frac{1}{2\pi RC}$$

The amplifier must satisfy the Barkhausen criterion:

$$A = 1 + \frac{R_f}{R_g} = 3$$

Without diode stabilization, the amplitude grows until clipping. With diodes, the feedback gain is automatically reduced, stabilizing the amplitude.

Part 1: Wien-Bridge Oscillator Without Diodes

Circuit Photo

Measured Waveform

Observation

Without diode stabilization, the gain is slightly above 3, so the amplitude increases until the op-amp output clips at the supply rails. This produces a square-like waveform.

Part 2: Wien-Bridge Oscillator With Diodes

Circuit Photo

Measured Waveform

Additional Waveform Capture

Observation

With diodes in the feedback path, the gain is automatically adjusted. As amplitude increases, diode conduction reduces gain slightly below 3, stabilizing the output and preventing clipping.

Calculations

Given:

$$R = 1\text{ k}\Omega, \quad R_f = 3.3\text{ k}\Omega, \quad R_g = 1\text{ k}\Omega, \quad C = 1\text{ }\mu\text{F}$$

Frequency:

$$f = \frac{1}{2\pi RC} = \frac{1}{2\pi(1000)(1 \times 10^{-6})} \approx 159.2\text{ Hz}$$

Gain:

$$A = 1 + \frac{R_f}{R_g} = 1 + \frac{3.3}{1} = 4.3$$

Ideal Wien-bridge requires $A = 3$. Thus the circuit without diodes will clip.

Output Measurement

From oscilloscope:

$$V_{peak} \approx 1.34\text{ V}$$

Conclusion

- The Wien-bridge oscillator without diodes produced a clipped waveform due to excessive gain.
- Adding diodes stabilized the amplitude, creating a clean sinusoidal output.
- Measured oscillation frequency was approximately 160 Hz, matching the theoretical value of 159.2 Hz.

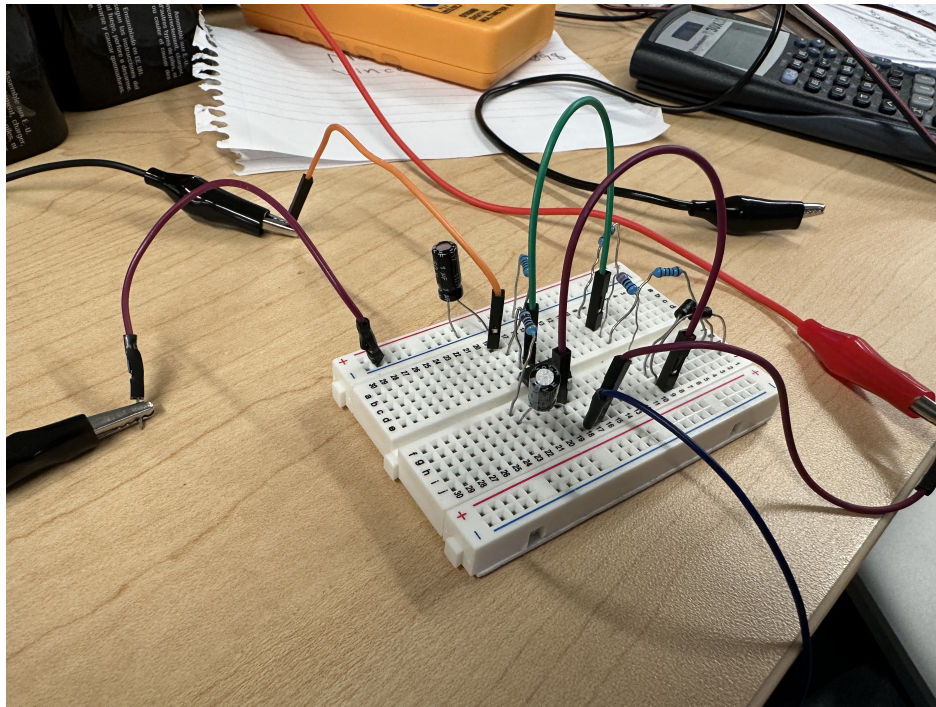


Figure 1: Breadboard setup for Wien-bridge oscillator without diodes

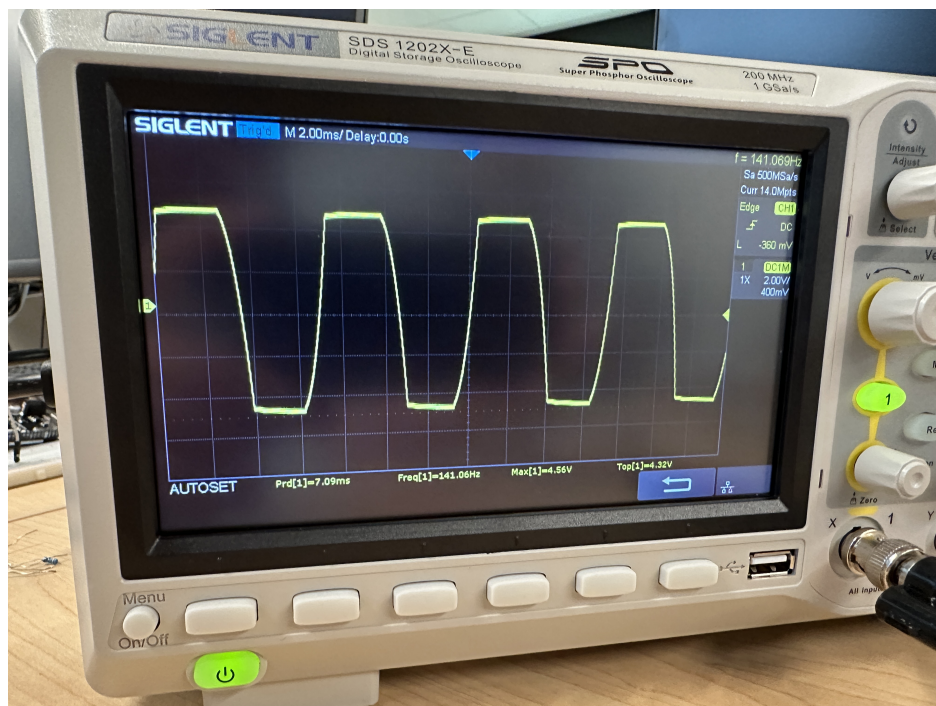


Figure 2: Oscilloscope waveform (clipped waveform due to no amplitude control)

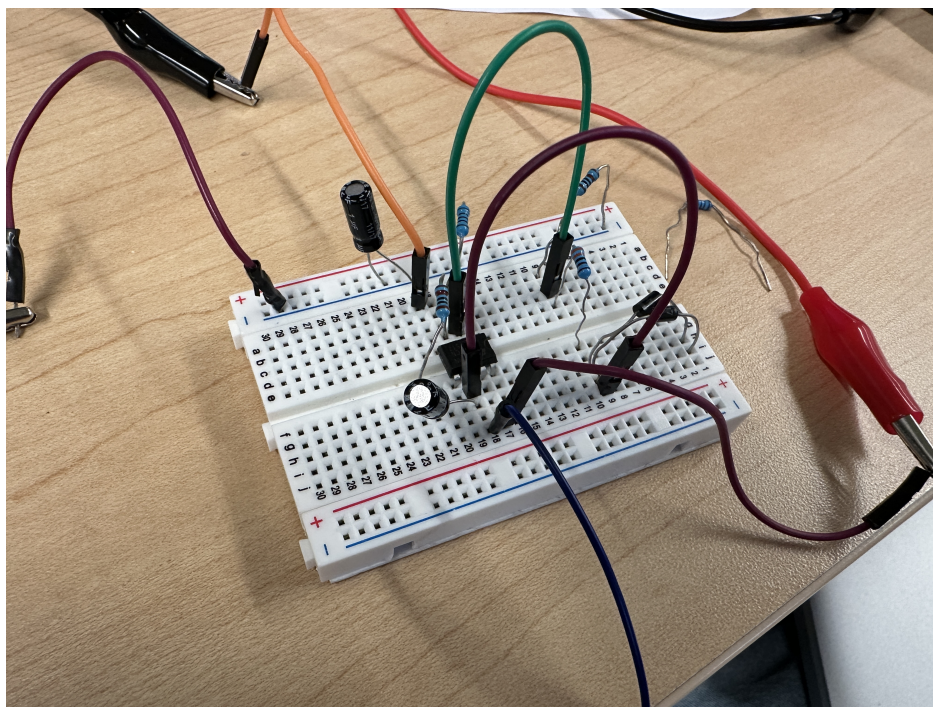


Figure 3: Breadboard setup with diodes for amplitude stabilization

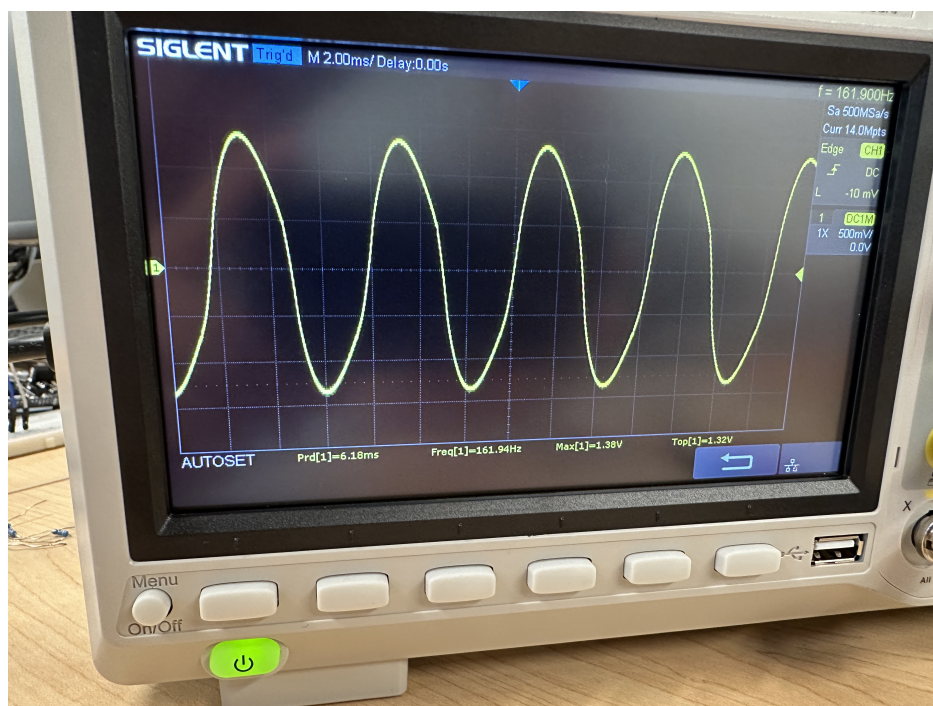


Figure 4: Oscilloscope waveform with diodes (cleaner sinusoid)